CPSC 121
Lab 03: Mathematical Calculations

For this lab, create a folder titled Lab_03 for VS Code. Inside this folder, you should have four classes, each representing the below programs. For example, you will have a class called ***SumOfCubes*** *which will host the solution to 1.1. The remaining classes should be titled,* ***Distance****,* ***VolumeSurfaceArea****, and finally,* ***AreaOfTriangle****.*

1.1: SumOfCubes

Write an application that prints the sum of cubes. Prompt for and read two integer values (use the **`Scanner`** class) and print the sum of each value raised to the third power (use the appropriate **Math** method). The result should be printed as an integer.

$$Sum\ of\ Cubes = num_1{}^3 + num_2{}^3$$

1.2: Distance

Write an application that prompts and reads (use the **`Scanner`** class) the (x, y) coordinates for two points as doubles. Compute and print the distance between the two points using the following formula (use the appropriate **Math** methods). Print the output to two decimal places using the **`DecimalFormat`** class.

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1.3: VolumeSurfaceArea

Write an application that prompts and reads the radius of a sphere (use the **`Scanner`** class) and prints its volume and surface area. Use the following formulas, in which *r* represents the spheres radius (use the appropriate **Math** methods). Print the output to four decimal places using the **`DecimalFormat`** class. (Note: if r = 3, volume and surface area should be equal)

$$Volume = \frac{4}{3}\pi r^3$$
$$Surface\ area = 4\pi r^2$$

1.4: AreaOfTriangle

Write an application that prompts and reads the lengths of the sides of a triangle from the user (use the Scanner class). Compute the area of the triangle using Heron's formula (below), in which *s* represents half of the perimeter of the triangle, and *a, b,* and *c* represent the lengths of the three sides (use the appropriate **Math** methods). **Note**: **DO NOT ask the user for the value of "s"; calculate it yourself**. Print the area to three decimal places using the **`DecimalFormat`** class.

$$Area = \sqrt{(s)(s-a)(s-b)(s-c)}$$

Rubric

- Does it work as described? (Needs my sign-off): **10 points**
- Did you follow style and coding practices and follow directions: **10 points**
- **Total: 20 points**

Submitting the Lab

Compress the submission files (see below) into a .zip file named: **Lab03_*LastName_FirstName*.zip** .
When you are done and ready to submit, submit the following files in the .zip file:

- `SumOfCubes.java`
- `Distance.java`
- `VolumeSurfaceArea.java`
- `AreaOfTriangle.java`